Adaptive Cross-Scene Foreground Segmentation via Learned Disputed-Pixel Arbitration

Sery Gahouidi Junior Thierry[1] and Zhe Chen[1*]

[1]College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, 211106, China.

*Corresponding author(s). E-mail(s): zhechen@nuaa.edu.cn;
Contributing authors: juniorsery00@gmail.com;

Abstract

Cross-scene video foreground segmentation via supervised–unsupervised model communication couples two complementary, imperfect sources of evidence: an unsupervised background model that adapts per scene but drifts on complex dynamics, and a supervised network accurate on its training distribution but poor on unseen scenes. The model-communication framework of Liang et al. combines the two, but resolves the pixels on which they disagree with a single fixed, hand-designed rule. We argue that this fixed resolution of the *disputed* pixels is the framework's central rigidity, and attach a lightweight, context-aware arbiter that learns to decide, from evidence it already computes, whether to trust the unsupervised model, the guidance, or the framework's own resolution. Unlike prior fusion, which either applies a *fixed* rule to such conflicts or *learns* only a frame-wide combination of homogeneous unsupervised detectors, ours learns the decision on the heterogeneous disputed set between a supervised and an unsupervised branch. To separate learning from reinforcement learning specifically, we also train a capacity-matched supervised control. On CDnet2014, the learned arbiter improves the re-implemented framework across a full factorial of two CPB representations (grayscale and RGB) and two guidance backbones (DeepLabV3+ and PSPNet); the benefit holds across all four of these configurations. A controlled ablation shows the reinforcement-learning agent significantly outperforms the supervised control in all four cells. In zero-shot transfer to LIMU the learned policy generalises to unseen scenes where a fixed rule cannot, and we characterise where an already-strong guidance branch resists it. It is plug-and-play, adding a few thousand parameters while leaving both models unchanged.

Keywords: Foreground segmentation, Background subtraction, Model communication, Learned arbitration, Reinforcement learning, Cross-scene generalisation

1 Introduction

Foreground segmentation underpins many vision systems, from surveillance to autonomous driving. It is intrinsically difficult because the appearance of the background is scene-dependent: illumination change, dynamic textures (water, foliage), camera jitter, and intermittent motion all break simple statistical models. Two paradigms dominate. Unsupervised background subtraction adapts online and needs no annotation but accumulates errors on complex dynamics; supervised semantic networks are accurate on their training distribution yet degrade on unseen scenes and require dense labels.

Liang et al. [1] proposed an elegant model-communication hybrid: a supervised guidance model (DeepLabV3+ [3] or PSPNet [4]) supplies a coarse prior that seeds selective updates for an unsupervised Co-occurrence Pixel-Block (CPB) model [2], which then refines the foreground through a three-stage communication procedure (a block-selection step, a block-swap step applied where the branches disagree, and a foreground-resemblance test). The framework is data-efficient and cross-scene, but its decision engine is a static heuristic: the CPB foreground test compares a weighted vote against a fixed relevance cutoff $\lambda = 0.5$, and disagreements between the branches are resolved by a hand-designed support-block rule. A single static rule cannot be optimal across the enormous variety of local contexts in a cross-scene benchmark.

Our thesis is that this arbitration should be *learned*, and that it need only be learned where the two branches actually disagree, the disputed pixels, which are typically only a small fraction of valid pixels (measured per scene, roughly 1.5–8.5% on cameraJitter and 1.7–20.6% on the hardest turbulence scenes, and below 10% on average). We attach a lightweight, context-aware arbiter to exactly those pixels and leave the rest of the framework untouched, keeping the method plug-and-play and computationally negligible while concentrating learning where the original framework is uncertain. We further ask which learning paradigm the arbitration requires, and design controlled experiments that separate the effect of *learned, context-aware* arbitration from the effect of reinforcement learning specifically.

This framing sharpens how our contribution differs from existing fusion. Semantic background subtraction and the Liang et al. communication framework already *localise* the disagreement between a supervised and an unsupervised branch, but resolve it with *fixed*, hand-designed thresholds and precedence rules [1, 15]. Conversely, learned fusers train a network to combine the outputs of several *homogeneous* unsupervised detectors over the *entire* frame, rather than arbitrating between two heterogeneous sources of evidence [16]. Among the methods we surveyed, including mixture-of-experts and selective-prediction approaches (Sec. 2.3), we are not aware of prior work that *learns* the decision on precisely the heterogeneous disputed set where a supervised guidance branch and an unsupervised background model conflict which is exactly the gap this paper fills, while the recent trend of ever-stronger foundation-model guidance [31, 36, 37] only makes the question of *how* to arbitrate between such complementary branches more pressing.

In light of the above, this paper makes the following contributions. First, we formulate disputed-pixel arbitration as a one-step contextual decision, with a 23-dimensional state summarising the evidence the framework already produces, a

three-way action space (*trust-CPB*, *trust-guidance*, or *keep-baseline*), and a segment-level soft-F_1 reward, and we learn it with the REINFORCE policy-gradient estimator augmented by a behaviour-cloning auxiliary; the keep-baseline action preserves the framework's own Stage-2/3 resolution, so the arbiter never discards the support-block machinery but only overrides it when confident. Second, we contribute a careful, honest re-implementation of the base framework as the controlled foundation for these comparisons, documenting the reconstruction choices we found necessary and reporting transparently where our reproduction still falls short of the authors' absolute level (Sec. 4.2). Third, on CDnet2014 and under a strict 5-fold scene-disjoint cross-validation protocol (the scenes partitioned into five folds, with four folds used for training and the held-out fold for testing) in which every system is scored on identical held-out pixels, we show that learned arbitration consistently improves the re-implemented framework across a full 2×2 of two unsupervised CPB representations (grayscale and RGB) and two guidance backbones (DeepLabV3+ and PSPNet): in the primary RGB / DeepLabV3+ configuration it raises the category-mean F-measure from 0.540 to 0.660 (+0.120) with all 11 categories improved, and, crucially, the reinforcement-learning agent *beats the capacity-matched supervised control* in a single-harness head-to-head (0.660 vs. 0.625 category-mean). This RL-over-supervised advantage is now *statistically significant in every one of the four configurations* (paired Wilcoxon over the 53 held-out scenes, p from 5×10^{-5} to 4×10^{-4}), with a strikingly stable category-mean margin of $+0.037 \pm 0.002$. The arbiter is *modular*, and the gain is *base-invariant across the four configurations tested*: the same improvement, and the same RL-over-supervised ordering, hold for each CPB representation and guidance backbone, with the arbiter retrained within each configuration. On the aggregation the original authors report for CDnet (pooled), our full RGB system under DeepLabV3+ guidance reaches 0.775, *comparable to* their reported 0.753; we give this only as a reference point, *not* a claim of superiority over the original method, and we build on a weaker reproduced CPB (on the stricter category-mean we remain below their computable level, which we state plainly in Sec. 4.2). Finally, through a set of controlled analyses, the single-harness head-to-head, a cross-dataset transfer to the LIMU dataset across the same 2×2, and a guidance-backbone ablation, we locate the source of the improvement in the *learned* nature of the arbitration and characterise its boundary: the agent beats the supervised control in transfer in all four configurations and is the best deployable system under DeepLabV3+, whereas under a PSPNet backbone that is already strong on the target dataset, simply trusting the guidance becomes hard for any arbiter to beat. Taken together, these results demonstrate that it is both possible and beneficial to learn the disputed-pixel arbitration of a supervised–unsupervised communication framework, without retraining either underlying model and while leaving the large majority of pixels (typically more than 90%) on which the two branches agree entirely unchanged.

The remainder of this paper is organised as follows. Section 2 reviews related work on background subtraction, learned decision fusion, and reinforcement learning for visual decision making. Section 3 recalls the base communication framework and presents the proposed arbiter, its state, action space, and reward. Section 4 describes the experimental protocol and reports the main result together with the controlled

analyses, and includes a dedicated discussion of the reproduction gap. Section 5 concludes.

2 Related Work

2.1 Unsupervised Background Subtraction

Unsupervised background subtraction methods model the appearance of the scene from video observations and determine foreground regions from deviations from the learned background. Classical approaches include parametric pixel-wise models, such as the adaptive Gaussian mixture model of Stauffer and Grimson [9], as well as non-parametric sample-consensus methods such as ViBe [10] and SuBSENSE [11]. These approaches have the important advantage of adapting to scene changes without requiring manually annotated foreground masks. However, their online adaptation can also accumulate erroneous observations, particularly in the presence of dynamic textures, intermittent motion, illumination changes, and other non-stationary background phenomena.

A different family of background models exploits spatial relationships rather than treating each pixel independently. The Co-occurrence Pixel-Block (CPB) model [2] represents a target pixel using a set of spatially correlated supporting blocks and detects foreground when the learned intensity relationship between the target and its supporting background blocks is violated. This spatial representation provides robustness to local appearance variation and does not require the type of continuously scheduled model update used by many conventional background subtraction methods. Nevertheless, once its statistical relationships have been established, CPB has limited ability to adapt when the selected supporting blocks become inappropriate for a changing scene, for example when a foreground object repeatedly occupies a region previously treated as background. The subsequent model-communication framework of Liang et al. [1] addresses this rigidity by using an external supervised model to guide the selection and replacement of CPB supporting blocks. Our work builds directly on this framework and targets what remains the governing rigidity of the combined system: how the disagreements between its two complementary sources of evidence are resolved.

2.2 Deep and Semantic Segmentation for Foreground Detection

Deep learning has substantially advanced foreground segmentation by learning spatial and, in some cases, temporal representations directly from annotated video data. CNN-based background subtraction methods such as DeepBS [12] and FgSegNet [13] demonstrated that learned representations can achieve very strong foreground segmentation accuracy, although often under a scene-specific protocol in which the network is exposed to frames of the target scene. Later work such as BSUV-Net [14] and its video-agnostic successor BSUV-Net 2.0 [33] explicitly addressed the harder setting of background subtraction for *unseen* videos, and more recent scene-independent methods push cross-scene generalisation further without any target-scene labels: the

zero-shot ZBS [34] models the background at the instance level using an open-vocabulary detector, while MUSTAN [35] exploits multi-scale temporal context to improve out-of-domain foreground segmentation. Nevertheless, CNN-based methods remain dependent on the availability and representativeness of training data, and their performance can deteriorate when the target scene exhibits characteristics that are poorly represented in the training distribution. In particular, the cross-scene setting is challenging because background appearance, camera behaviour, illumination, and object configurations can vary substantially between scenes.

Semantic segmentation provides another useful source of high-level visual information. Networks such as PSPNet [4] and DeepLabV3+ [3] learn object-level representations and can provide semantic priors even when foreground-specific annotations are unavailable. When trained on datasets such as ADE20K [8], these networks can identify classes that are commonly associated with foreground objects. More recently, promptable foundation segmenters such as the Segment Anything Model [31] provide class-agnostic masks of strikingly high spatial quality, and test-time adaptation methods [22, 32] can adjust a segmentation network to a target distribution online; both are natural, stronger drop-in replacements for the guidance branch used here. A fast-growing line of very recent work indeed builds moving-object segmentation directly on such foundation models by coupling them with motion cues [36, 37], confirming that stronger guidance is increasingly available off the shelf and that the question of *how* to combine a powerful semantic prior with a motion/background cue is more pertinent than ever. We deliberately adopt a fixed, publicly available ADE20K prior rather than a foundation-model prior, both to remain a faithful reproduction of Liang et al. and because our contribution, learning the arbitration between the two branches, is orthogonal to, and would only benefit from, a stronger guidance source. However, semantic segmentation is not equivalent to motion-based foreground detection: semantic networks primarily describe *what* an image region represents, whereas background subtraction must additionally determine *whether* that region constitutes foreground in the current scene and temporal context. Consequently, semantic predictions can contain both false positives and false negatives when object identity alone is insufficient to distinguish a moving foreground object from a static background object.

Several studies have therefore investigated combinations of conventional background subtraction and semantic information. Semantic background subtraction methods combine a background model with a semantic branch through predefined decision mechanisms [15, 16]. Liang et al. [1] substantially generalised this idea by introducing a bidirectional communication framework in which a supervised segmentation model provides coarse guidance for the unsupervised CPB model, while CPB subsequently contributes fine-grained information to the final segmentation. Their framework lets the guidance prediction steer which background blocks are chosen and when they are swapped, and then recovers foreground regions by a resemblance test in feature space. This combination is particularly relevant to our work because it establishes a complementary relationship between semantic and statistical evidence rather than requiring either branch to solve foreground segmentation independently. However, the interaction between the branches remains governed by predefined rules.

Our work retains this complementary architecture but replaces its fixed arbitration on ambiguous pixels with a learned decision mechanism.

2.3 Learned Decision Fusion and Expert Routing

The problem of deciding which of several competing predictors should be trusted has been studied extensively in ensemble learning and mixture-of-experts models. Mixtures of experts introduce a learned gating mechanism that assigns different inputs to specialised predictors [17], while sparsely gated mixture-of-experts architectures further scale this principle by dynamically selecting a subset of experts for each input [18]. Related work has also investigated probabilistic fusion of multiple background classifiers, allowing complementary classifiers to contribute jointly to the final background decision [19]. These approaches establish an important general principle: when different models possess complementary strengths, a learned decision mechanism can determine which source should dominate for a particular input.

Our problem is related to this general notion of learned gating, but differs in both scope and purpose. The proposed arbiter is not a generic ensemble combiner that independently aggregates several complete segmentation networks. Instead, it is embedded inside the CPB–guidance communication framework of Liang et al. [1] and is invoked specifically where the two existing branches disagree. The decision mechanism therefore operates on a restricted set of ambiguous pixels rather than replacing either segmentation model. Moreover, its input is derived from evidence already produced by the framework, including CPB confidence and support statistics, guidance information, and local scene context. Finally, the arbiter is trained with the objective of improving the resulting segmentation quality rather than merely learning a generic classifier combination rule. The proposed formulation can therefore be viewed as a specialised, framework-aware gating mechanism whose purpose is to resolve disagreement between two heterogeneous foreground cues.

2.4 Cross-Scene Generalisation and Domain Adaptation

Generalising visual segmentation models from observed scenes to unseen environments is closely related to domain adaptation and domain generalisation. Methods such as AdaptSegNet [20] and CyCADA [21] address domain discrepancies by learning representations or output spaces that are more invariant between source and target domains. Test-time adaptation methods such as Tent [22] instead adapt a pretrained model during inference, typically using information from the target distribution itself. Meta-learning and few-shot segmentation approaches [23, 24] provide another strategy by learning models that can rapidly adapt to a new task or scene from a small amount of target information.

These approaches differ from the cross-scene setting considered here in an important respect. The proposed arbiter is trained using source scenes and then applied to previously unseen scenes without requiring target-scene annotations, target-scene fine-tuning, or an online adaptation stage. The underlying CPB model remains scene-specific through its normal background modelling process, while the learned arbiter captures a more general decision rule for resolving disagreement between CPB and

the supervised guidance model. Furthermore, the arbiter does not attempt to transform the entire feature distribution of the target scene; it operates only on pixels for which the two existing segmentation sources provide conflicting evidence. This restricted intervention reduces the amount of learned behaviour that must generalise across domains and allows the original communication framework to remain unchanged elsewhere.

2.5 Reinforcement Learning for Visual Decision Making

Reinforcement learning has previously been used to formulate visual processing as a sequence of decisions rather than as a single feed-forward prediction. Active object localization with deep reinforcement learning [25], for example, learns an agent that iteratively transforms a candidate bounding box to localise an object. Similarly, ADNet [26] formulates visual tracking as a sequence of learned actions, while Han et al. [6] introduced a reinforcement-learning-based cutting agent for video object segmentation. The latter is particularly relevant to the present work because it demonstrates how reinforcement learning can be used to make localised decisions that progressively influence visual segmentation.

Our formulation differs from these sequential visual decision problems. The proposed agent does not search for an object, move a bounding box, or modify a region through a long sequence of actions. Instead, at each disputed pixel, it observes the local evidence produced by the existing segmentation framework and chooses which available decision source should be trusted. Consequently, the fundamental learning problem is substantially more local: each disputed pixel corresponds to a one-step contextual decision, with no action-induced transition that changes the state of another disputed pixel. This makes the formulation closer to a contextual bandit, or equivalently a one-step Markov decision process, than to a long-horizon sequential MDP. The distinction is important when selecting the learning algorithm: we therefore employ the policy-gradient estimator REINFORCE [7], rather than methods such as PPO [27] or SAC [28], which were developed for general sequential decision problems with longer horizons and explicit temporal credit assignment. The formulation allows the learned policy to focus specifically on the ambiguous cases generated by model communication while leaving pixels on which the two branches agree unchanged.

Overall, the literature reveals a progression from independent statistical background modelling, to supervised and semantic segmentation, and subsequently to hybrid systems that combine these complementary sources of information. Liang et al. [1] provide the closest foundation to our work by demonstrating that supervised guidance can compensate for the limitations of CPB while CPB can in turn refine the coarse supervised prediction. However, their communication mechanism ultimately relies on fixed decision rules. Existing mixture-of-experts and fusion methods demonstrate the value of learned gating, and reinforcement-learning studies establish the feasibility of learned visual decision making, but neither line of work specifically addresses cross-scene, per-pixel arbitration between an unsupervised CPB model and a supervised guidance model within the Liang communication framework. Our work addresses this gap by learning a lightweight, context-aware arbiter that intervenes only on the small subset of pixels where the two branches disagree, thereby preserving

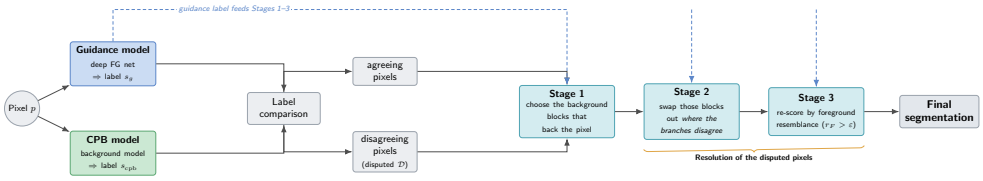


Fig. 1 The base cross-scene segmentation framework of Liang et al. [1] that we reproduce, drawn at the level of a single pixel p. The pixel receives a supervised guidance label s_g and an unsupervised CPB label s_{cpb}; a label comparison sorts pixels into *agreeing* and *disagreeing* (disputed) sets. All pixels then pass through the three-stage model communication, Stage 1 (choosing the background blocks that back the pixel), Stage 2 (swapping those blocks out *where the two branches disagree*), and Stage 3 (re-scoring the pixel by foreground resemblance, $r_F > \varepsilon$), with the guidance label feeding every stage. Stages 2–3 are the fixed hand-designed resolution of the disputed pixels $\mathcal{D}$, applied by a rule that is the same for every scene.

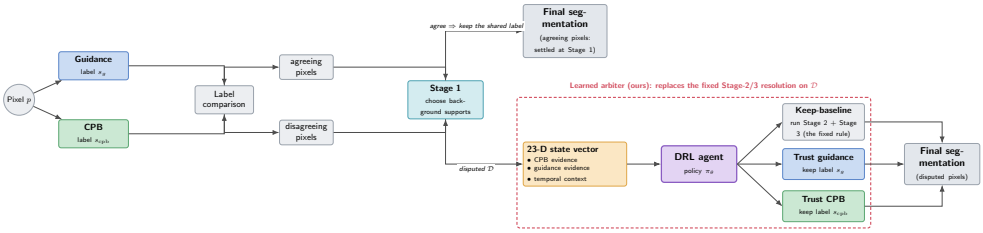


Fig. 2 Our method at the level of a single pixel. The perception is identical to Fig. 1: guidance and CPB label the pixel and a comparison sorts it. Pixels on which the two branches *agree* keep the shared label (settled at Stage 1). For a *disputed* pixel we build a 23-D state vector, CPB evidence, guidance evidence and temporal context, all quantities the framework already computes, and a lightweight DRL policy π_θ chooses one of three actions: *keep-baseline* (run the original Stage-2/3 fixed rule), *trust-guidance* (keep s_g), or *trust-CPB* (keep s_{cpb}). Only the fixed Stage-2/3 resolution of $\mathcal{D}$ is replaced (dashed box); the perception of both branches and the agreeing pixels are untouched.

the original communication pipeline while making its critical arbitration step adaptive rather than fixed.

3 Method

3.1 Framework recap and the dispute set

We adopt the CPB + guidance + model-communication framework of [1, 2] without modifying its perception. For a target pixel p with supporting background blocks $\{Q_m^B\}_{m=1}^M$ ranked by Pearson correlation, CPB scores foregroundness by a weighted vote over the "unsteady" supports and compares it to a fraction of the total weight:

$$\sum_{m=1}^{M} \gamma_m \, \omega_m \;>\; \lambda \sum_{m=1}^{M} \gamma_m, \tag{1}$$

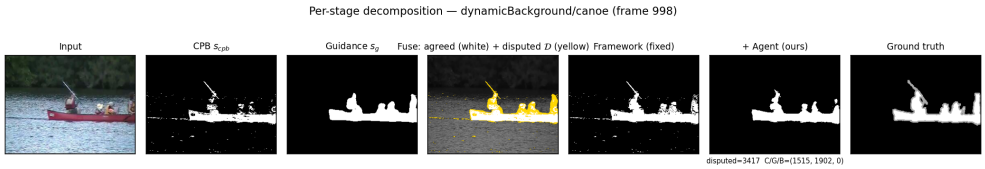


Fig. 3 Per-stage decomposition on a real frame (*dynamicBackground/canoe*, frame 998, RGB CPB, DeepLabV3+ guidance, held-out fold). From left to right: the input; the unsupervised CPB label s_{cpb}, which recovers the boat but also over-fires across the rippling water (a classic dynamic-background failure); the supervised guidance s_g, which recovers the canoe and paddlers cleanly with almost no water response; the *Fuse* view, which marks on a dimmed input the *agreed foreground* (white, where both branches say foreground) and the *disputed set* $\mathcal{D}$ (yellow, where the two branches disagree, the only pixels the agent acts on, here 3417 of them, concentrated on the boat boundary, the paddle, and scattered water speckles); the fixed framework s_{fix}, which carries water false positives through; the framework with the learned agent, which suppresses the water and tightens the mask to the boat; and the ground truth. On this frame the agent's action mix on $\mathcal{D}$ is trust-CPB 1515 / trust-guidance 1902 / keep-baseline 0. Only pixels in $\mathcal{D}$ are ever changed; the agreed pixels and both branches' perception are untouched.

where γ_m is the correlation weight of support m, $\omega_m \in \{0, 1\}$ marks whether its intensity relation to p is broken, and $\lambda = 0.5$ is the fixed relevance cutoff. Let $s_{\text{cpb}}(p)$ and $s_g(p)$ be the CPB Stage-1 and guidance labels. The dispute set is $\mathcal{D} = \{p : s_{\text{cpb}}(p) \neq s_g(p)\}$. Agreed pixels are decided by the original three-stage rule, giving the framework's own label $s_{\text{fix}}(p)$; our method changes the decision only on $\mathcal{D}$ (Fig. 3).

Importantly, on a disputed pixel $s_{\text{fix}}(p)$ is the output of the *complete* communication procedure, the Stage-2 block swap followed by the Stage-3 resemblance test, computed exactly as in [1]. Because our keep-baseline action (below) selects s_{fix}, the arbiter never discards the framework's support-block machinery; it only decides, per disputed pixel, whether to accept that machinery's output or to override it with a raw branch label. All pixels the Stage-2/3 machinery modifies lie within $\mathcal{D}$, so the arbiter and the original communication act on exactly the same set.

3.2 Arbitration as a one-step contextual decision

For each disputed pixel the arbiter observes a context s_t and selects one action a_t; the action determines that pixel's label and does not alter the context of any other pixel. There is thus no state transition induced by the action, so the process is a contextual bandit (equivalently, a one-step MDP), not a multi-step MDP. The reward for a frame is computed from the counts induced by all the frame's actions simultaneously (below), so it is invariant to the order in which pixels are processed. This framing is why a policy-gradient estimator without temporal bootstrapping is the appropriate learner.

State $s_t \in \mathbb{R}^{23}$**.**

The state summarises the local evidence around a disputed pixel using only quantities the framework already computes, in three groups (Table 1): CPB evidence (the continuous score $\sum \gamma_m \omega_m / \sum \gamma_m$, live-support count, a short temporal window of the CPB score including a persistence statistic, and local spatial statistics including connected-component blob size), guidance evidence (probability, confidence, and

Table 1 The 23-dimensional arbitration state s_t. All features derive from quantities the framework already produces; no new perception is added.

group	features	intuition
CPB evidence (11)	score, margin, live-support count, 1-step Δscore, window mean/std/slope/persistence, spatial-context mean/std, connected-component blob size	foregroundness and confidence of the unsupervised vote; object vs. flicker
Guidance (7)	label, probability, confidence, soft disagreement, distance-in/out, local density, distance-to-confident-region	strength and geometry of the semantic prior
Context (5)	motion, edge, texture, contrast, guidance reliability	generic scene cues

geometry relative to confident guidance regions), and generic context (motion, edge, texture, contrast).

Action $a_t \in \{0, 1, 2\}$***.***

Three discrete actions resolve the dispute (Fig. 2): 0 = trust CPB (s_{cpb}), 1 = trust the guidance (s_g), 2 = keep the framework's own decision (s_{fix}). Action 2 everywhere reproduces the original framework exactly, so the policy departs from the reference only when it expects to help.

Reward.

Because the F-measure is non-differentiable in the discrete per-pixel decisions, we optimise it directly through the reward. For a frame with running counts (TP, FP, FN) under the agent's mask we use the marginal contribution of each pixel to the scene soft-F_1,

$$w_{tp} = \frac{2(FP+FN)}{(2TP+FP+FN)^2}, \quad w_{pen} = \frac{2\,TP}{(2TP+FP+FN)^2}, \tag{2}$$

assigning $+w_{tp}$ to a true positive and $-w_{pen}$ to a false positive/negative (normalised), plus small shaping terms that reward fixes (correcting a pixel the static rule got wrong), penalise breaks (harming a pixel it got right), and add a confidence term. The reward is masked by the CDnet region of interest, and the advantage is centred by *this frame's* own mean reward (a per-frame baseline), which removes a bias a global moving-average baseline induces on low-reward scenes. Because this baseline is the mean over the frame's disputed pixels, any single action's influence on it is $O(1/|\mathcal{D}|)$ and negligible for the disputed-set sizes here; it therefore acts as a variance-reduction control variate rather than a strictly action-independent baseline.

Policy, training, and the behaviour-cloning auxiliary.

The policy π_θ is a compact network ($23 \to 64 \to 64 \to 3$) trained with REINFORCE. On each dispute exactly one primitive side matches the ground truth; we add a behaviour-cloning (BC) auxiliary that supervises the policy toward that side, whose weight is linearly annealed from 1.0 down to 0.2 over training. The BC signal injects the (largely

Algorithm 1 Training the disputed-pixel arbiter (REINFORCE with a behaviour-cloning auxiliary).

Require: training frames; policy π_θ; BC weight schedule β ($1.0 \to 0.2$); learning rate η
Ensure: trained policy π_θ

for each training iteration **do**
  sample a frame; run CPB and guidance to obtain labels s_{cpb}, s_g
  $\mathcal{D} \leftarrow \{\, p : s_{\text{cpb}}(p) \neq s_g(p) \,\}$ ▷ disputed set
  for all $p \in \mathcal{D}$ **do**
    $x_p \leftarrow \text{state}(p)$ ▷ 23-dimensional feature vector
    sample $a_p \sim \pi_\theta(\cdot \mid x_p)$, $a_p \in \{\text{trust-CPB, trust-guidance, keep-baseline}\}$
  end for
  form the frame mask: apply $\{a_p\}$ on $\mathcal{D}$, keep the shared label on agreed pixels
  for all $p \in \mathcal{D}$ **do**
    $r_p \leftarrow$ marginal contribution of p to the frame soft-F_1 ▷ $+w_{tp}$ / $-w_{pen}$, Eq. (2), plus shaping
  end for
  $b \leftarrow \frac{1}{|\mathcal{D}|} \sum_{p \in \mathcal{D}} r_p$ ▷ per-frame advantage baseline
  $a_p^\star \leftarrow$ oracle action for each $p \in \mathcal{D}$ ▷ the side matching the ground truth
  $g \leftarrow \sum_{p \in \mathcal{D}} (r_p - b)\, \nabla_\theta \log \pi_\theta(a_p \mid x_p) \;+\; \beta \sum_{p \in \mathcal{D}} \nabla_\theta \log \pi_\theta(a_p^\star \mid x_p)$
  $\theta \leftarrow \theta + \eta\, g$; anneal β
end for

separable) arbitration cue that pure REINFORCE extracts slowly, while the reward retains the freedom to defer to the framework where the rule inverts. A patch-CNN policy variant that additionally reads a local window of three maps was implemented but *disabled* for the reported agent because it was slower to train and did not improve accuracy; the agent therefore uses the same scalar inputs as the supervised control; checkpoints are selected by a harm-aware criterion (mean category gain penalised by the worst category). Algorithm 1 summarises the full training loop.

3.3 A three-way supervised arbiter as control

To test whether the *learned nature*, rather than reinforcement learning specifically, drives the gain, we train a capacity-matched supervised classifier on the identical signal and the identical *three-way* action space {trust-CPB, trust-guidance, keep-baseline}. Its target is the oracle action per disputed pixel: keep-baseline whenever the framework's own decision already matches the ground truth, otherwise the correct primitive branch. Sharing the agent's action set, and in particular the keep-baseline action, is what makes the supervised-vs-RL comparison fair: a two-way control lacking keep-baseline would be forced to overwrite every disputed pixel with a raw branch, discarding the framework's own reliable resolution on the many disputes where it is already correct. The control shares the policy's inputs, effective action set, and training scenes, differing only in the learning paradigm.

4 Experiments

4.1 Datasets, protocol, and metrics

We evaluate primarily on CDnet2014 (11 categories, 53 scenes) and study cross-dataset transfer on LIMU. CPB models are trained per scene on the first 400 frames; the guidance is DeepLabV3+ with public ADE20K weights (PSPNet is used for the backbone ablation), not fine-tuned on the evaluation data. The arbiters are evaluated with 5-fold scene-disjoint cross-validation: the 53 scenes are partitioned into five folds, and the policy/classifier is trained on four folds and tested on the held-out fold, so *no test scene is seen in training* and every scene is tested exactly once across the folds. (A full leave-one-scene-out protocol would require 53 folds, which is computationally impractical on our CPU-only setup; the five-fold grouping preserves the no-leakage property at a fraction of the cost.) We report three aggregations: category-mean (macro over the 11 categories, the official CDnet metric, which is our primary internal number), scene-mean (macro over scenes), and pooled (micro-average over all pixel counts). Matching the reference's aggregation per table is essential to an honest comparison. The exact aggregation Liang et al. use for their "Overall" column is not stated in their paper, so we reconstruct it from their own per-category cells: their published CDnet "Overall" (0.753 for CPB + DeepLabV3+, 0.772 for PSPNet) does *not* equal the arithmetic mean of their 11 reported categories (that computable category-mean is 0.771), but is consistent with a *pooled* micro-average over pixel counts; their LIMU "Overall" (≈ 0.793), by contrast, is consistent with the *scene-mean* over the three LIMU scenes. For a reference numerical comparison we therefore report pooled F-measure alongside their CDnet number and scene-mean alongside their LIMU number, while the standard category-mean remains our *primary* metric. Because the exact aggregation formula underlying their published numbers is not stated, these comparisons to Liang et al. should be read as reference points rather than as a claim that our evaluation exactly reproduces their scoring protocol; our load-bearing claims are the controlled, identical-pixel deltas within our own pipeline, which are independent of this ambiguity. This distinction is not cosmetic: on LIMU it can invert the sign of a comparison (Sec. 4.5). Following CDnet, hard shadow (50) counts as background and unknown regions (85/170) are excluded. Key parameters follow Liang et al.: $\eta = 2.5$, $\sigma = 0.8$, REF = 20, block-background threshold = 7 foreground pixels of 64, $\lambda = 0.5$, with *no* post-processing.

On aggregation and cross-validation.

Because our arbiter is a *cross-scene learned* component, cross-validation is required: the arbiter must be tested on scenes it did not train on. This differs from the base framework of [1], which has no cross-scene learned component (its CPB is trained per scene), and therefore reports all scenes directly without folds. We report the full-CV category-mean over all scenes; we do *not* report a best-fold number, which would be a form of test-set selection.

Implementation details (for reproducibility).

CPB models are trained per scene on the first 400 frames and then frozen; the foreground test uses the full-block (all-64) steady/unsteady statistic and the paper-faithful Stage-2 replacement, with $\eta = 2.5$, $\sigma = 0.8$, REF = 20, a block-background threshold of 7 foreground pixels of 64, a per-support scale floor of 6, $\lambda = 0.5$, and no post-processing; the grayscale and RGB variants differ only in the number of CPB channels. Guidance is a public DeepLabV3+ (or PSPNet for the backbone study) with ADE20K weights, not fine-tuned; the soft guidance probability is the sum of the twelve ADE20K foreground-class channels. The arbiter is a scalar three-way policy ($23 \to 64 \to 64 \to 3$, the patch-CNN branch disabled) trained with REINFORCE under a per-frame advantage baseline and a segment-level soft-F_1 reward (fix/break/confidence shaping weights 0.5/0.5/0.05), plus a behaviour-cloning auxiliary linearly annealed from 1.0 to 0.2; optimisation uses Adam at learning rate 3×10^{-4}, 8 frames per update, gradient-norm clipping at 1.0, entropy annealed from 0.01 to 0.003, a keep-baseline bias init of 1.5, and 20,000 steps per fold. Checkpoints are selected by a harm-aware criterion (mean category gain penalised by the worst category) on a validation set carved one scene per category from the training folds. The capacity-matched supervised control shares the identical 23-D input, three-way action space, and training folds, and is trained by class-balanced cross-entropy on the per-pixel oracle action. Evaluation uses a 60-frame uniform sample within each scene's temporal ROI, scored at native ground-truth resolution; we validated this sampling against full-sequence scoring (the CPB category-mean over the 60-frame sample matches the all-frames value to within 0.003), so the sample is representative of the whole scene. All runs use a fixed seed (1234) and the same 5-fold scene-disjoint partition. Experiments run on CPU (PyTorch, OpenCV, MMSegmentation); we will release the code, configurations, and trained checkpoints. In terms of cost, the arbiter is a $23 \to 64 \to 64 \to 3$ MLP ($\approx$ 6k parameters) evaluated by a single forward pass on only the disputed pixels ($< 10\%$ of the frame on average); its inference cost is therefore negligible next to the CPB and guidance passes it arbitrates, and it adds no parameters to either underlying model.

4.2 The reproduction gap, in detail

Because our contribution is measured *relative* to a reproduced base framework, the fidelity of that reproduction deserves an honest discussion. In short: the gap to the authors' published numbers is component-level, lies mainly in the unsupervised CPB rather than in the communication logic, and is closed by the learned arbiter on the authors' own metric.

Fidelity.

Two details most affect fidelity. First, the CPB steady/unsteady test: our reconstruction compares the target pixel against the *entire* 8×8 support block (satisfied only when all 64 pixels agree), our reading of the matrix-valued condition in [1], whose code is not public. An earlier block-*average* port is far more permissive and floods dynamic-background and turbulence scenes with false positives; restoring the full-block rule recovers the authors' qualitative behaviour (on LIMU *LightSwitch* the framework rises from ≈ 0.42 to ≈ 0.78, matching their 0.769). Second, parameterisation: we use Liang

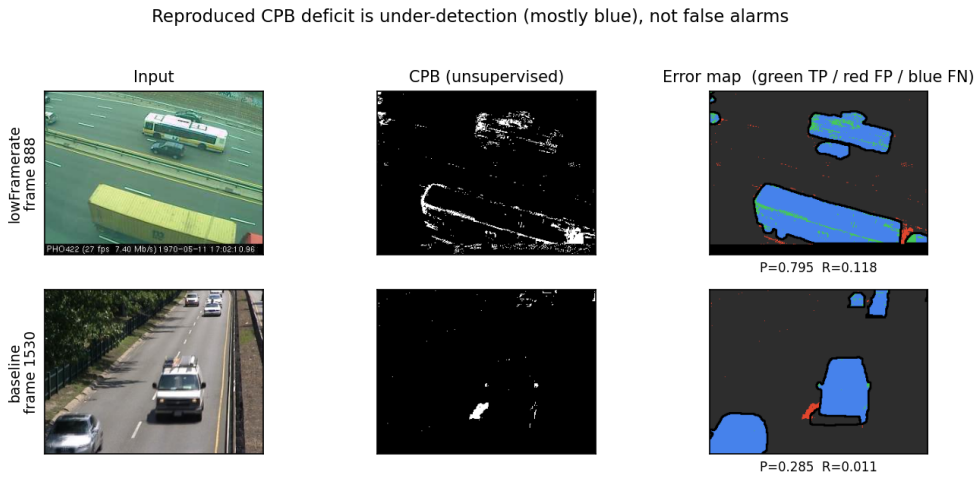


Fig. 4 The reproduced CPB deficit is *under-detection*, not false alarms. Per-frame error maps of the pure unsupervised CPB for two hard scenes (green = true positive, red = false positive, blue = false negative). Almost all of the lost score is *blue*, large, slow, or dim foreground that the frozen CPB *misses* while red (spurious over-firing) is negligible. On *lowFramerate/turnpike* the recall collapses to 0.118 while precision stays moderate (0.795); on *baseline/highway* the frozen CPB misses the foreground almost entirely (recall 0.011). In both cases the dominance of blue over red is the telltale signature of an inflated per-support scale σ: the acceptance band $\eta\sigma$ is too wide and genuine foreground is absorbed into the background model, rather than the model over-firing. This is the mechanism diagnosed and partially corrected below.

et al.'s settings ($\eta = 2.5$, $\sigma = 0.8$, REF = 20, block-background threshold 7/64) with *no* post-processing, so the reproduced system is theirs and not a variant.

The gap is component-level, mainly on the CPB side.

Comparing on the authors' aggregation (pooled F-measure), our faithful framework reaches 0.57 versus their 0.753. The *guidance* is close ($\approx$ 0.62 vs. $\approx$ 0.68; in fact stronger than theirs on cameraJitter, thermal and badWeather, its deficits confined to turbulence and intermittent motion, intrinsic limits of single-frame segmentation). The dominant deficit is the *CPB itself* ($\approx$ 0.42, RGB 0.426 measured, vs. 0.63); our metric harness is verified by reproducing standard baselines from the authors' own rows (GMM 0.570 vs. 0.571; SuBSENSE 0.738 vs. 0.741). A weak CPB caps the whole pipeline, which is why the authors' bare CPB (0.63) can exceed our full framework (0.57). Crucially, the learned arbiter closes this on the authors' metric: our full RGB / DeepLabV3+ system reaches pooled 0.775 > 0.753. We do not overstate this, on the stricter category-mean we remain below their computable 0.771 (the weaker public guidance and structurally hard categories such as turbulence and PTZ), but on the aggregation they report, the component-level shortfall does not carry through to the delivered system.

Table 2 Diagnosing and partially closing the reproduction gap (CDnet2014, category-mean F-measure, all 53 scenes, 30-frame protocol). The deficit is on the CPB side; a robust σ estimator and a guidance-reliability gate recover ≈one third of the gap without touching the guidance or the communication logic. We nonetheless adopt the *faithful base* for all controlled comparisons and report these as a validated diagnosis; the learned arbiter recovers more (+0.120 category-mean in the primary configuration) than both hand-engineered fixes combined (+0.076).

configuration	category-mean F-measure
CPB alone (ours)	0.374
guidance alone (ours)	0.480
re-implemented framework (faithful base)	0.513
+ robust σ (median/MAD)	0.540
+ robust σ + guidance-reliability gate	**0.589**
Liang et al. (CPB + DeepLabV3+)[†]	0.771

[†]For an apples-to-apples comparison in this category-mean table we list the *computable* category-mean 0.771 (the arithmetic mean of the authors' 11 published per-category CDnet cells). Their reported "Overall" of 0.753 is instead a pooled micro-average (see Sec. 4.1); the pooled comparison against 0.753 is made in Table 3. The gate has a single hyper-parameter (frame guidance-coverage $< 0.1\%$ disables the Stage-2 deletion); results are stable over $[0.02\%, 0.1\%]$. Its residual failures are scenes with accurate-but-sparse guidance, which coverage cannot distinguish from empty guidance, the case the learned arbiter resolves from confidence instead. This recoverability diagnostic uses a lighter 30-frame, all-scenes pass; the main experiments (Sec. 4.3) use the 60-frame, 5-fold scene-disjoint protocol, under which the same faithful base scores 0.540.

Diagnosing and partially closing the CPB deficit.

The deficit is one of *under-detection*: the reproduced CPB is consistently low-recall, and the error maps (Fig. 4) show the lost score is almost entirely missed foreground, not spurious responses. The cause is a too-tolerant background model: the per-support scale σ is estimated by a plain standard deviation over training frames that on busy scenes already contain moving objects, so the foreground spikes inflate σ, the acceptance band $\eta\sigma$ widens, and genuine foreground is absorbed, exactly the "training-sample quality" effect the authors warn about. That their bare CPB still reaches 0.633 suggests practical estimation choices the paper does not spell out (a robust σ, or cleaner foreground-free training frames); our from-the-equations reproduction is faithful to the text but not to those unstated details. Two targeted fixes confirm the diagnosis (Table 2): a robust median/MAD σ estimator raises the framework category-mean from 0.513 to 0.540, and a guidance-reliability gate (which suppresses the Stage-2 deletion of correct CPB detections when the guidance is near-empty) raises it further to 0.589 recovering about one third of the gap without touching the guidance or the communication logic.

Why this reinforces, rather than undercuts, the contribution.

Two points make the gap supportive of our claim. First, our effect is measured *relative* to the reproduced base, inside the same pipeline and on *identical* disputed pixels, so any deficiency in the absolute CPB or guidance level is present equally in the baseline and the arbitrated system and cancels in the difference; the +0.11 to +0.14

category-mean improvement is thus invariant to the reproduction level. Second, the learned arbiter recovers *more* than the two hand-engineered fixes combined (+0.120 vs. +0.076), without any hand-set threshold: the fixed reliability gate must judge guidance trustworthiness from a single coverage number and fails on accurate-but-sparse guidance (e.g. *shadow/cubicle*), whereas the arbiter resolves the same decision per pixel from CPB confidence and local context. The fixes are a brittle, hand-crafted instance of what the arbiter learns adaptively; we therefore adopt the faithful base for all comparisons and take the +0.120 improvement (Sec. 4.3) as our load-bearing claim.

4.3 Main result: learned vs. static arbitration on CDnet2014

Table 3 is the paper's anchor, and Fig. 5 visualises its per-category structure. Every system shares the same CPB model and the same guidance; only the resolution of disputed pixels differs. All five systems are scored in a *single* harness under 5-fold scene-disjoint cross-validation, so every number in the table is computed on identical held-out pixels. We report the primary RGB / DeepLabV3+ configuration here and give the full 2×2 over representations and backbones in Sec. 4.7 (Table 7, Fig. 8). Replacing the static rule with the learned arbiter raises the category-mean F-measure from 0.540 to 0.660 (+0.120), with *all* 11 *categories improved*. We stress that this static baseline (0.540) is our *faithful* reproduction (Sec. 4.2) scored under the main 5-fold, 60-frame protocol, *not* the robust-σ variant of Table 2; the same faithful base reads 0.513 there only because that table uses a 30-frame, all-scenes diagnostic protocol, and the robust-σ and gate rows of Table 2 are a separate recoverability analysis that enters *no* controlled comparison. The improvement is also robust across aggregations, the agent exceeds the static framework on pooled (0.775 vs. 0.567, +0.208), scene-mean (0.667 vs. 0.543, +0.124) and category-mean (+0.120) alike, and the per-fold category-mean gain measured in the cross-validation is $+0.138 \pm 0.070$, i.e. consistent across folds rather than driven by a single lucky split. Notably the fixed rule barely helps here: on this configuration the static framework's pooled F-measure (0.567) is *below* guidance-alone (0.616), so almost the entire delivered gain is the arbiter's. On the pooled aggregation the authors report, our full system reaches 0.775, comparable to their 0.753; we present this only as a reference point, *not* a superiority claim, since the exact aggregation is unspecified and our stricter category-mean remains below theirs (Sec. 4.2). The learned arbiter's action mix on held-out scenes is roughly trust-CPB 0.45, trust-guidance 0.45, keep-baseline 0.10: it exercises all three actions, and in particular the keep-baseline action is genuinely used. The *oracle* row is the ceiling of this exact decision problem: on each disputed pixel it picks the one of the three actions (trust-CPB, trust-guidance, keep-baseline) that matches the ground truth, while all agreed pixels are frozen at their shared label, so it upper- bounds what any disputed-pixel arbiter can achieve, and the agreed-pixel errors it cannot touch are exactly why its ceiling is 0.825 rather than 1.0.

Notes on Table 3. All rows come from the single scoring harness, which trains the supervised control on the exact scenes each fold's agent saw and scores every system on the same held-out pixels; the per-fold spread (±0.070) is measured in the DRL cross-validation. The spread reflects genuine cross-fold difficulty differences rather than one anomalous split, the five per-fold category-mean gains are +0.112, +0.146, +0.076,

Table 3 CDnet2014, 5-fold scene-disjoint CV, every scene tested once, identical held-out pixels, single scoring harness. Primary configuration: RGB CPB, all-64 framework, DeepLabV3+ guidance. Category-mean is the official CDnet metric; pooled is the aggregation the authors report as "Overall".

system	pooled	scene-mean	cat.-mean
guidance (DeepLabV3+)	0.616	0.540	0.533
framework (Liang, static)	0.567	0.543	0.540
framework + supervised MLP	0.750	0.628	0.625
framework + DRL agent (ours)	**0.775**	**0.667**	**0.660**
oracle (ceiling)	0.910	0.827	0.825
Liang et al. (CPB+DeepLabV3+), pooled	*0.753*	,	

agent − framework (cat.): +0.120; MLP − framework (cat.): +0.085; agent − MLP (cat.): +0.035 (significant, $p = 3.9 \times 10^{-4}$, Wilcoxon).
On pooled the full system (0.775) exceeds the authors' reported "Overall" (0.753).
Per-fold agent category-mean gain: $+0.138 \pm 0.070$ (5 folds), 11/11 categories improved.

Table 4 Per-category CDnet2014 category-mean F-measure (5-fold, every scene once, single harness), primary RGB / DeepLabV3+ configuration: framework vs. supervised MLP vs. DRL agent. Sorted by the agent's gain over the framework. The agent improves the framework in 11/11 categories and beats the supervised MLP in 9/11.

category	framework	+MLP	+agent	agent−fw
dynamicBackground	0.612	0.693	0.842	+0.230
PTZ	0.135	0.403	0.359	+0.224
shadow	0.719	0.911	0.926	+0.207
intermittentObjectMotion	0.526	0.626	0.683	+0.157
baseline	0.650	0.744	0.801	+0.151
badWeather	0.607	0.720	0.742	+0.135
lowFramerate	0.494	0.538	0.568	+0.074
turbulence	0.484	0.567	0.540	+0.056
cameraJitter	0.727	0.681	0.763	+0.036
nightVideos	0.323	0.324	0.356	+0.033
thermal	0.665	0.664	0.678	+0.013
category-mean	**0.540**	**0.625**	**0.660**	**+0.120**

+0.084, and +0.270, i.e. *every* fold shows a clear positive gain (the smallest is +0.076, none is near zero), with the wide standard deviation driven by one unusually strong fold rather than by any fold failing. More importantly, the per-scene test removes the fold as the unit of analysis: the per-scene agent−framework gain is strongly significant (paired Wilcoxon over the 53 scenes, $p \approx 1.0 \times 10^{-7}$), as is the agent−MLP gain ($p = 3.9 \times 10^{-4}$); both learned arbiters improve the framework at $p < 10^{-5}$. The full significance picture across all four configurations is in Table 8.

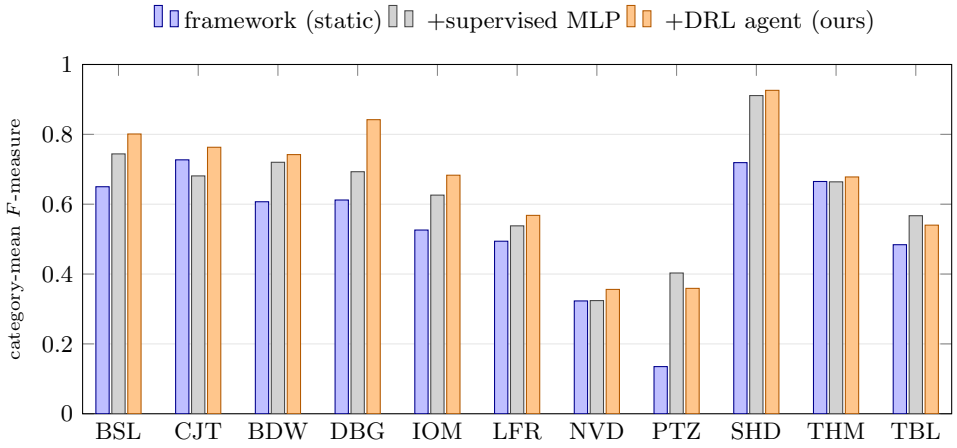


Fig. 5 Per-category CDnet2014 category-mean F-measure (5-fold scene-disjoint CV, identical held-out pixels), primary RGB / DeepLabV3+ configuration from the single-harness head-to-head. Category codes: BSL baseline, CJT cameraJitter, BDW badWeather, DBG dynamicBackground, IOM intermittentObjectMotion, LFR lowFramerate, NVD nightVideos, PTZ, SHD shadow, THM thermal, TBL turbulence. The learned arbiter (orange) improves the static framework in *all* 11 categories and beats the supervised MLP in 9/11; the two exceptions (PTZ, turbulence) are the categories where the guidance-heavy MLP happens to help most. On *DBG*, where the branches disagree the most, the agent adds the largest gain (+0.230).

Notably the static framework's pooled F-measure (0.567) is below guidance-alone (0.616): on this configuration the fixed rule does not reliably help, and the arbiter's role is to recover the scenes it collapses. The gain concentrates exactly where the static rule is weakest (see the per-category breakdown in Fig. 5 and Table 4). Fig. 6 shows representative qualitative results, and Fig. 7 contrasts the fixed rule's and the learned arbiter's *final* masks on a single scene together with their pixelwise difference.

4.4 Supervised-vs-RL: a single-harness head-to-head

The natural question is whether the improvement requires reinforcement learning or merely a learned, context-aware decision. To answer it rigorously, rather than by comparing two separately-run protocols, we evaluate both arbiters in a *single* scoring harness that trains the three-way supervised MLP on the exact scenes each fold's agent saw, gives it the identical 23-D signal and the identical three-way action space, and scores both on the same held-out pixels. On this common harness in the primary configuration (Table 3) the supervised MLP reaches category-mean 0.625 (+0.085 over the static framework) and the DRL agent reaches 0.660 (+0.120); the agent therefore beats the capacity-matched supervised control by +0.035 category-mean. The ordering is consistent across aggregations, the agent also leads on pooled (0.775 vs 0.750) and scene-mean (0.667 vs 0.628), and it holds in 9 of the 11 categories, with the agent's largest margins over the MLP on exactly the categories the MLP mishandles: *dynamicBackground* (+0.149), *cameraJitter* (+0.082) and *baseline* (+0.057). Relative to the oracle ceiling on disputed pixels (0.825), the agent recovers 42% of the available head-room and the MLP 30%.

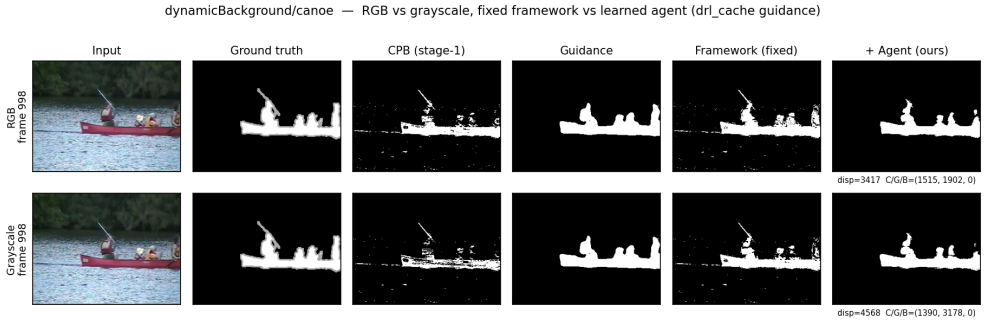


Fig. 6 Qualitative comparison on *dynamicBackground/canoe* (frame 998, held-out fold, DeepLabV3+ guidance), for both CPB representations (top: RGB, bottom: grayscale). Columns: input, ground truth, CPB Stage-1, guidance, the fixed framework s_{fix}, and the framework with the learned agent. The unsupervised CPB over-fires across the rippling water (a classic dynamic-background failure), and the fixed framework carries much of that water response through, heavily so in grayscale. The learned agent, acting only on the disputed pixels, suppresses the water and tightens the mask to the boat and paddlers, visibly closer to the ground truth. The effect is the same under both representations, illustrating the base-invariance of the contribution; the agent uses both primitive actions here (trust-CPB and trust-guidance) and never resorts to a no-op.

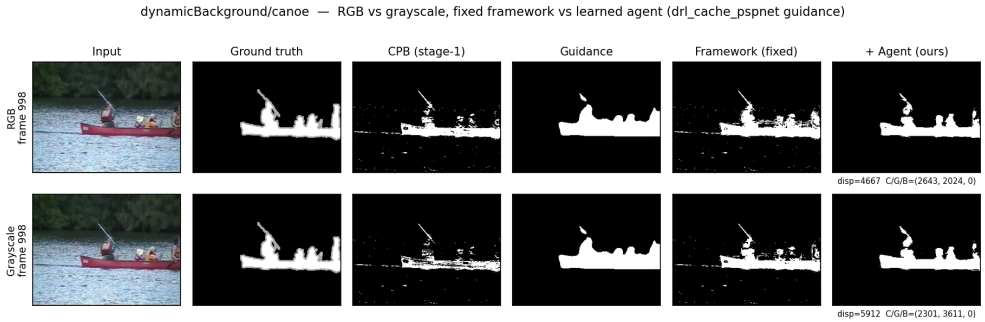


Fig. 7 The same scene and frame under a *different* guidance backbone (PSPNet), columns as in Fig. 6. Although the PSPNet guidance mask differs in shape from DeepLabV3+, the learned agent again removes the framework's water false positives and recovers a clean boat under both CPB representations, the arbiter's cleanup is plug-and-play across the backbone, not tuned to one guidance network's particular errors. Only disputed pixels are ever changed; the agreed majority is left untouched.

Crucially, and unlike an earlier single-configuration run, this RL-over-supervised advantage is now *statistically significant, and significant in every one of the four configurations* of the 2×2 (paired Wilcoxon over the 53 held-out scenes; Table 8): $p = 3.9 \times 10^{-4}$ (RGB / DeepLabV3+), 1.2×10^{-4} (RGB / PSPNet), 2.2×10^{-4} (grayscale / DeepLabV3+) and 5.3×10^{-5} (grayscale / PSPNet), with the agent winning on 37–41 of the 53 scenes in each case. The category-mean margin is remarkably stable at $+0.037 \pm 0.002$ across the four cells. The learned-versus-static effect is of course even stronger, both learned arbiters improve the framework at $p < 10^{-5}$, and the

agent−framework gain reaches $p \approx 10^{-7}$. We therefore state the RL-over-supervised result as a firm, reproducible finding rather than a point estimate, while scoping it precisely to what we tested: *under the matched three-way action architecture and the identical* 23*-dimensional input*, the REINFORCE arbiter consistently and significantly outperforms the capacity-matched supervised MLP control across all four CPB representation × guidance backbone configurations. We do not claim RL is universally superior to supervised learning for arbitration; we claim that in this controlled, matched setting it is, and that the effect is stable across the pipeline variants we evaluated.

Why reinforcement learning helps here.

The advantage has a concrete mechanistic cause rather than being a tuning artefact. The *per-pixel optimal* action on disputes, the supervised target, is keep-baseline 60–70% of the time, trust-CPB 22–33%, and trust-guidance only 6–9%; the MLP, trained to match those labels, is correspondingly cautious. The RL agent instead learns a much more decisive policy (roughly trust-CPB 0.45, trust-guidance 0.45, keep-baseline 0.10) and still wins, because it is optimised on the *segment-level* soft-F_1 reward rather than on per-pixel action accuracy. Decisively trusting a branch across a disputed region can raise the segment F_1 even where it disagrees with the per-pixel optimal action on individual pixels; the MLP optimises the wrong surrogate (per-pixel action correctness) while the agent optimises the metric the paper actually reports. The clearest illustration is *dynamicBackground*, the category where the two branches disagree the most: the per-pixel-trained MLP is too cautious and reaches only 0.693, whereas the reward-aware agent commits to the correct branch across the disputed water and reaches 0.842, beating the MLP by +0.149 there. (In the grayscale configuration, where the static CPB on this category is stronger still, the same mechanism is even starker: the MLP actively *collapses* the category while the agent limits the loss, the representation changes the numbers but not the RL-over-supervised ordering.) We therefore frame the contribution precisely: the improvement comes from making the arbitration *learned and context-aware*, and reinforcement learning is significantly better than supervised imitation at it, attractive because it optimises the non-differentiable segment metric directly, an advantage that now holds significantly across both representations and both backbones.

4.5 Cross-dataset transfer to LIMU

The deployment-realistic test of cross-scene ability is to train on one dataset and apply the arbiter unchanged to another. We train a single general arbiter (and a single general supervised control) on *all* CDnet scenes and evaluate it zero-shot on LIMU (CameraParameter, Intersection, LightSwitch), with CPB trained per scene on LIMU as in [1]. We report *scene-mean* F-measure, the aggregation the reference uses for its LIMU "Overall" (≈ 0.793); this choice matters, as we show below. Table 5 gives the DeepLabV3+ transfer for both CPB representations; the arbiters are applied to LIMU without any adaptation.

Three observations follow, each stated separately.

Table 5 Cross-dataset zero-shot transfer to LIMU under *DeepLabV3+* guidance, both CPB representations, scene-mean F-measure (the reference's LIMU convention ≈ 0.793). A single CDnet-trained arbiter/control, applied unchanged. Under this backbone the agent is the best deployable system: it beats the base, the supervised control *and* raw guidance.

config	base	guidance	supervised MLP	**DRL agent**	oracle
RGB CPB	0.486	0.599	0.486	**0.617**	0.886
grayscale CPB	0.445	0.599	0.528	**0.604**	0.876

Zero-shot: the CDnet-trained agent and MLP are applied to LIMU with no re-training. agent−MLP = +0.131 (RGB) and +0.077 (grayscale); agent−guidance = +0.018 (RGB) and +0.006 (grayscale). In RGB the supervised control degenerates to keep-baseline almost everywhere and scores at the base level (0.486), whereas the agent keeps a consistent CPB/guidance mix; this erratic-supervised-vs-consistent-agent behaviour is the recurring transfer pattern.

The agent is the best deployable system under DeepLabV3+. The ordering is monotone base < guidance < MLP < agent < oracle in grayscale, and in RGB the supervised control degenerates so that base = MLP < guidance < agent. In both, the agent is the only non-oracle system to beat even raw guidance (+0.018 RGB, +0.006 grayscale), purely from a policy learned on a different dataset.

The agent beats the supervised control off-distribution. Central to the thesis, the agent leads the capacity-matched MLP by +0.131 (RGB) and +0.077 (grayscale) scene-mean on *identical* disputed pixels, the value of the learned policy over supervised imitation of the oracle. Under domain shift the supervised control is erratic (in RGB it collapses to a full keep-baseline no-op and simply returns the base) while the agent's behaviour stays consistent.

The aggregation matters. On the grayscale configuration, *pooled* F-measure would make the supervised control appear to win (agent−MLP = −0.072), but on *scene-mean*, the reference's own LIMU metric, the comparison reverses to +0.077 in the agent's favour. Reporting the metric the reference actually uses is what keeps the comparison honest.

4.6 Cross-dataset transfer under a different guidance backbone (PSPNet)

DeepLabV3+ is our primary guidance backbone, and the transfer above is the main cross-dataset result. As a robustness check we repeated the zero-shot LIMU transfer with PSPNet guidance (agent and MLP retrained on CDnet under PSPNet, then applied to LIMU unchanged) for both CPB representations, and the outcome is worth stating plainly rather than burying. PSPNet guidance *itself* transfers strongly to LIMU (scene-mean F-measure 0.703, well above DeepLabV3+ guidance's 0.599), so on this small three-scene target the best simple policy becomes "trust the guidance", and the agent (Table 6) does not beat guidance-alone (0.670 RGB, 0.684 grayscale vs. 0.703). The benefit of the learned arbiter is therefore *conditional*: it is realised when the guidance branch is the unreliable one in the target domain, as with DeepLabV3+ above. What does *not* change under PSPNet is the RL-versus-supervised picture: the agent still beats the capacity-matched supervised control in both representations (+0.070 RGB, +0.180 grayscale) and still improves the base. It is the "arbiter beats

Table 6 Cross-dataset transfer under *PSPNet* guidance, both CPB representations, scene-mean F-measure. PSPNet guidance itself transfers strongly (0.703), so trusting it is already the best simple policy; the agent improves the base and beats the supervised control in both representations, but cannot exceed near-ceiling guidance, an honest boundary of the method.

config	base	guidance	supervised MLP	DRL agent	oracle
RGB CPB	0.471	**0.703**	0.600	0.670	0.868
grayscale CPB	0.430	**0.703**	0.504	0.684	0.851

All LIMU figures are scene-mean F-measure (the reference's LIMU "Overall" convention). agent−MLP = +0.070 (RGB), +0.180 (grayscale); agent−guidance = −0.033 (RGB), −0.019 (grayscale). The supervised control remains erratic under shift (heavy keep-baseline use), while the agent keeps a consistent guidance-leaning mix.

strong guidance" claim, not the "learned beats fixed" or "RL beats supervised" claims, that does not hold on this near-ceiling backbone.

The shortfall against guidance is localised and diagnosable rather than a random degradation. Having learned on CDnet, where the base is weak almost everywhere, to arbitrate purely between CPB and guidance, the agent converges to a degenerate keep-baseline rate (≈ 0.00) and has no effective way to *defer* to a base or guidance decision that happens to be excellent on the target. The keep-baseline action that was harmlessly unused under DeepLabV3+ becomes the missing tool under a backbone strong enough to make trusting the guidance worth it, pointing to a transfer-time prior that revives keep-baseline when both branches are locally confident (Sec. 4.8). Even so, that the RL-over-supervised margin is *largest* here in grayscale (+0.180) underlines that the agent's advantage over supervised imitation is most valuable exactly when the arbitration is hardest.

4.7 Plug-and-play robustness: the full 2 × 2

A learned arbiter could, in principle, merely memorise the error pattern of one CPB representation or one guidance network. To test that it learns to arbitrate heterogeneous disagreement rather than a single pipeline's particular mistakes, we run the entire in-domain CDnet head-to-head over the full 2 × 2 of two CPB representations (grayscale, RGB) and two guidance backbones (DeepLabV3+, PSPNet), each under 5-fold cross-validation in the single harness. Table 7 reports all four cells in the three aggregations, and Fig. 8 visualises the category-mean. The result is uniform: in every cell the systems are strictly ordered in category-mean, framework < supervised MLP < DRL agent < oracle (guidance-alone is *not* uniformly below the framework: on the weaker cells it can exceed it, e.g. in pooled F-measure of the primary cell, so we do not place it in this ordering); the agent improves the framework by +0.11 to +0.14 category-mean and beats the capacity-matched supervised control, *significantly in all four* (Table 8). The margin over the supervised control is strikingly constant at $+0.037 \pm 0.002$ category-mean regardless of representation or backbone. We stress that the arbiter is retrained within each configuration's cross-validation; what this establishes is therefore that the *method* improves the framework across all

Table 7 Full 2×2 on CDnet2014 (in-domain, 5-fold, identical held-out pixels): the framework, the three-way supervised MLP, and the learned agent under two CPB representations and two guidance backbones. Category-mean is the official metric; scene-mean and pooled are given for completeness. The agent leads the MLP, which leads the framework, in *every* cell.

configuration	metric	framework	+MLP	+agent
RGB / DeepLabV3+	cat.-mean	0.540	0.625	**0.660**
	scene-mean	0.543	0.628	**0.667**
	pooled	0.567	0.750	**0.775**
RGB / PSPNet	cat.-mean	0.532	0.632	**0.668**
	scene-mean	0.531	0.631	**0.671**
	pooled	0.536	0.743	**0.760**
Gray / DeepLabV3+	cat.-mean	0.527	0.599	**0.639**
	scene-mean	0.528	0.602	**0.642**
	pooled	0.654	0.742	**0.757**
Gray / PSPNet	cat.-mean	0.524	0.615	**0.652**
	scene-mean	0.522	0.614	**0.654**
	pooled	0.621	0.736	**0.770**

A base/metric trade-off is visible, grayscale gives higher pooled, RGB higher category-mean, but the *gain* is invariant, which is the point. On pooled (the authors' reported aggregation) both DeepLabV3+ cells reach or exceed their level (0.775, 0.757 vs. 0.753); both PSPNet cells are at parity (0.760, 0.770 vs. 0.772).

four configurations without modifying the underlying models, *not* that one already-trained policy can be dropped, untrained, onto a different base. The only place a trained policy is reused unchanged is the cross-*dataset* transfer to LIMU (Sec. 4.5).

4.8 Limitations

Four limitations are worth stating plainly. First, our re-implemented base framework is below the authors' absolute level (Sec. 4.2); we mitigate this by making every comparison a controlled, identical-pixel delta, but the absolute numbers should be read as a re-implementation rather than a state-of-the-art claim. Second, while the agent improves all 11 categories in the primary RGB / DeepLabV3+ configuration, it can still slightly harm an isolated category in a cell where the static rule is already strong, in grayscale / DeepLabV3+ that is *nightVideos* (-0.026), the sole regression of the four cells, because acting on disputes there risks overriding a CPB decision that was already correct; the keep-baseline action limits but does not eliminate this cost. Third, the benefit of the learned arbiter is conditional on the guidance branch being the unreliable one in the target domain. The PSPNet cross-dataset transfer (Sec. 4.6) makes this concrete: when PSPNet guidance transfers strongly to LIMU (scene-mean 0.703), trusting it is already the best simple policy and the CDnet-trained agent, though it still beats the supervised control and improves the base, falls below guidance-alone (0.670 RGB, 0.684 grayscale). The cause is that the agent, having converged to a degenerate

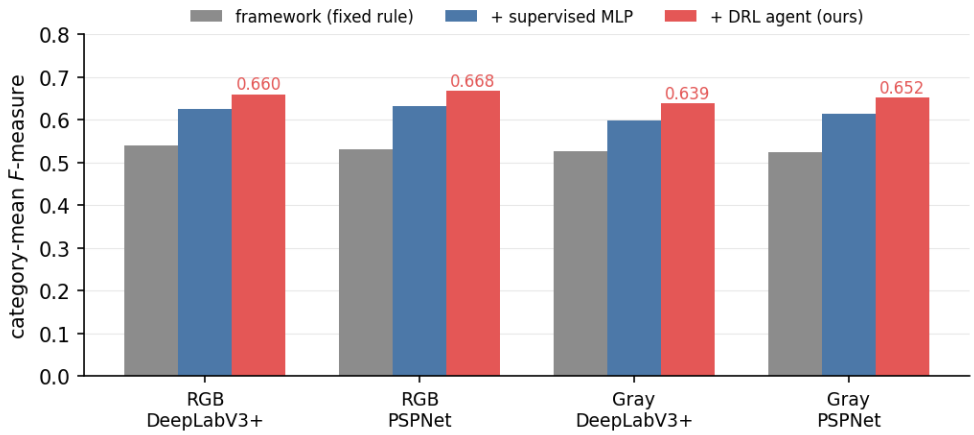


Fig. 8 Plug-and-play robustness: category-mean F-measure across the full 2×2 of CPB representation $\times$ guidance backbone. The learned agent (red) improves the fixed framework (grey) and the supervised MLP (blue) in every configuration, with a near-constant margin, evidence that the gain is a property of the learned arbitration, not of a single pipeline.

Table 8 Paired Wilcoxon significance over the 53 held-out scenes and per-fold gain, all four configurations. The RL agent beats the supervised control significantly everywhere, with a near-constant category-mean margin.

	agent vs. framework		agent vs. MLP	
configuration	Δcat	p	Δcat	p
RGB / DeepLabV3+	+0.120	1.0×10^{-7}	+0.035	3.9×10^{-4}
RGB / PSPNet	+0.136	1.6×10^{-7}	+0.036	1.2×10^{-4}
Gray / DeepLabV3+	+0.112	3.8×10^{-7}	+0.040	2.2×10^{-4}
Gray / PSPNet	+0.128	9.2×10^{-8}	+0.037	5.3×10^{-5}

All p-values are exact two-sided paired Wilcoxon over the 53 held-out scenes; the agent beats the framework on 42–45 of 53 scenes and the MLP on 37–41. Per-fold agent−framework gain (category-mean): RGB-DL +0.138±0.070; RGB-PSP +0.155±0.070; Gray-DL +0.126±0.042; Gray-PSP +0.141±0.046.

keep-baseline rate of ≈ 0.00 on CDnet cannot defer to a branch that happens to be excellent on the target. A transfer that re-enables keep-baseline when both branches are locally confident, or a small amount of target-domain calibration of the action prior, is the natural remedy and is left to future work. Fourth, all policies are trained with a single random seed (1234); our significance tests and the four-configuration study address scene and configuration variability, but not the variability of retraining the policy itself, so the RL-over-supervised margin should be read as established across scenes and configurations rather than across training randomness. A multi-seed study is left to future work.

5 Conclusion

We revisited the model-communication framework for cross-scene foreground segmentation and identified its static arbitration rule as the key rigidity. By learning a lightweight per-pixel policy that acts only on the small set of disputed pixels (typically well under 10%), and by giving that policy a keep-baseline action that preserves the framework's own Stage-2/3 resolution, we improved our re-implemented framework's category-mean F-measure on CDnet2014 by +0.11 to +0.14 across a full 2×2 of CPB representations and guidance backbones (primary RGB / DeepLabV3+: $0.540 \rightarrow 0.660$, all 11 categories improved), with a tight per-fold spread. As a reference comparison, on the pooled aggregation consistent with the authors' published "Overall" the full system reaches 0.775 versus their 0.753, though on the stricter category-mean it remains below their computable level, the delivered gain, not an absolute state-of-the-art claim, is our contribution. Controlled studies, a single-harness supervised-vs-RL head-to-head, cross-dataset transfer to LIMU, and a full base×backbone robustness sweep, place the contribution precisely: it is the *learned, context-aware* arbitration that matters, and, under the matched three-way action architecture and identical input, reinforcement learning is *significantly* better at it than a capacity-matched supervised control in all four configurations (p from 5×10^{-5} to 4×10^{-4}), with a near-constant +0.037 category-mean margin, so the arbiter is plug-and-play and robust across the representation and backbone variants we evaluated. The transfer study also delimits when the learned arbiter helps: under DeepLabV3+ guidance, which transfers weakly to LIMU, the agent is the best deployable system, whereas under a PSPNet backbone that itself transfers strongly it beats the supervised control but falls below guidance-alone the benefit is conditional on the guidance branch being the unreliable one in the target domain, and we trace the residual loss to a degenerate keep-baseline rate; whether a small transfer-time calibration can remedy this without target labels is left to future work rather than claimed here. The controlled methodology, faithful re-implementation, identical-pixel comparison, head-to-head, transfer, and backbone generalisation, offers a template for evaluating learned components inside classical vision pipelines. A particularly promising direction is to extend learned arbitration beyond the disputed set to the pixels on which the two branches *agree yet are jointly wrong*: the present method leaves all agreed pixels untouched, but a learned policy that also identifies and corrects confident-but-wrong agreements could recover errors that no disputed-pixel arbiter can reach. Further directions include reviving the keep-baseline action under strong backbones and extending the method to further guidance backbones and datasets. A temporal-consistency reward that couples decisions across frames is also promising: a genuinely sequential formulation could give reinforcement learning a structural edge over the one-step decision studied here.

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Consent for publication. Not applicable.

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Code availability. The source code, configurations, and trained checkpoints will be released on a public repository upon publication.

Author contributions. S. Gahouidi Junior Thierry: conceptualization, methodology, software, investigation, formal analysis, validation, visualization, and writing of the original draft. Z. Chen: supervision, conceptualization, and writing, review and editing. All authors read and approved the final manuscript.

Use of AI-assisted technologies. During the preparation of this work the authors used an AI-based assistant for language editing and formatting only. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the manuscript.

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